SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE CODE: **MLA03**
COURSE NAME: Reinforcement learning for Environment and Agent

---

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

I am A.Shiva Shankar Reddy(192425174) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
A.Shiva Shankar Reddy
Signature of the student:
QUESTIONS: 10 **Total Marks: 50**

Annexure A

Data Interpretation **Duration: 1hr**

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

ANS:

1. Identify the RL Concept

Reward Maximization and **Policy Improvement**.

2. Examine the Data

- Initial Reward = **10**
- Final Reward = **45**
- Reward increases over episodes.

3. Analyze the Results

The agent learns from experience and selects better actions in each episode, so the reward keeps increasing.

4. Interpret Using RL Principles

Higher rewards mean the agent is improving its policy and learning the best actions.

5. Compare the Outcomes

- Initial episodes: Low reward and poor performance.
- Later episodes: High reward and better performance.

6. Justify

The agent updates its knowledge after every episode, which improves decision-making and increases rewards.

7. Conclusion

The increasing reward shows that the agent is learning successfully and moving toward the optimal policy.

8. RL Terminology

Agent, Reward, Policy, Episode, Learning, Optimal Policy.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance.

ANS:

1. Identify the RL Concept

Exploration vs Exploitation.

2. Examine the Data

- $\varepsilon = 0.1 \rightarrow$ Highest reward.
- $\varepsilon = 0.3 \rightarrow$ Medium reward.
- $\varepsilon = 0.5 \rightarrow$ Lowest reward.

3. Analyze the Results

Lower epsilon means the agent uses the best learned action more often. Higher epsilon causes more random actions.

4. Interpret Using RL Principles

More exploitation gives better rewards, while too much exploration reduces performance.

5. Compare the Outcomes

Epsilon	Reward
0.1	Highest
0.3	Moderate
0.5	Lowest

6. Justify

At $\varepsilon = 0.1$, the agent already has good knowledge and mostly selects the best action, giving higher rewards.

7. Conclusion

$\varepsilon = 0.1$ gives the best performance because it balances exploration and exploitation effectively.

8. RL Terminology

Epsilon-Greedy, Exploration, Exploitation, Reward, Policy.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

ANS: **(5 Marks)**

1. Identify the RL Concept

Q-Learning vs SARSA.

2. Examine the Data

- Q-Learning gets higher rewards.
- SARSA gets slightly lower rewards.

3. Analyze the Results

Q-Learning learns faster because it always updates using the best possible future action.

4. Interpret Using RL Principles

Higher rewards show that Q-Learning finds the optimal policy faster than SARSA.

5. Compare the Outcomes

Algorithm	Performance
Q-Learning	Higher Reward
SARSA	Lower Reward

6. Justify

Q-Learning is an off-policy algorithm, so it learns the optimal action more quickly.

7. Conclusion

Q-Learning converges faster and performs better than SARSA in this case.

8. RL Terminology

Q-Learning, SARSA, Off-Policy, Reward, Convergence, Optimal Policy.

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

ANS:

1. Identify the RL Concept

Discount Factor (γ).

2. Examine the Data

- $\gamma = 0.2 \rightarrow$ Lower reward.
- $\gamma = 0.5 \rightarrow$ Medium reward.
- $\gamma = 0.9 \rightarrow$ Highest reward.

3. Analyze the Results

Higher discount factors help the agent consider future rewards, improving overall performance.

4. Interpret Using RL Principles

The agent learns long-term strategies instead of only immediate rewards.

5. Compare the Outcomes

Discount Factor	Performance
0.2	Low
0.5	Medium
0.9	High

6. Justify

A higher discount factor encourages better long-term decision-making.

7. Conclusion

$\gamma = 0.9$ provides the best learning performance.

8. RL Terminology

Discount Factor, Future Reward, Policy, Reward, Long-Term Learning.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

1. Identify the RL Concept
Policy Learning and **Success Rate**.
2. Examine the Data

- 50 Episodes → **45% Success**
- 200 Episodes → **92% Success**

3. Analyze the Results
The agent gains more experience with training and makes better decisions.
4. Interpret Using RL Principles
The higher success rate shows that the agent has learned an effective policy.
5. Compare the Outcomes

Episodes	**Success Rate**
50	45%
200	92%

6. Justify
More training allows the agent to improve its policy and reduce mistakes.
7. Conclusion
The agent has learned successfully and is close to the optimal policy.
8. RL Terminology
Success Rate, Agent, Policy, Episode, Learning, Optimal Policy.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand